SARVAJANIK COLLEGE OF ENGINEERING AND TECHNOLOGY

COMPUTER ENGINEERING DEPARTMENT

B. Tech III, SEM V, Div- II

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Open Ended Problem (OEP)

on

"ProjectHub — PROJECT MANAGEMENT WEBSITE"

Subject Name: Web Technologies (BTCO13502)

Prepared and Presented by Group No. : 5

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1. INTRODUCTION

1.1 Project Overview

ProjectHub is a web-based project management platform that simplifies project creation, organisation, tracking and team collaboration through a centralised workspace. It reduces scattered information and improves coordination by integrating project details, team records and progress tracking.

The application uses PHP, MySQL, HTML, CSS, Bootstrap and JavaScript. Its features include user registration and login, a statistical dashboard, project management, status and priority tracking, due-date validation, team-member management and a searchable directory. PHP sessions maintain login details, while MySQL stores project and user information. The responsive interface supports access through common web browsers.

1.2 Problem Statement

Managing multiple projects manually can create difficulties in maintaining accurate records, monitoring deadlines, coordinating team members and tracking the completion status of individual projects.

Challenges without a centralised system:

- Scattered project data.
- Difficult status and deadline tracking.
- Repetitive manual work.
- Slow team information searches.
- Poor record organisation.

ProjectHub centralises project data, simplifies collaboration and improves tracking.

1.3 Proposed Solution

ProjectHub centralises project and team management through a browser-based interface. The system includes:

1. User registration and login.
2. Dashboard with project statistics.
3. Project creation, editing, deletion and search.
4. Priority, status and deadline management.
5. Team-member management.
6. Searchable team directory.
7. Session-based access control.
8. Server-side validation and database operations.
9. Responsive Bootstrap-based interface.

1.4 Objectives

Objectives:

- Develop a web-based project management system.
- Implement user registration, login and sessions.
- Perform CRUD operations on projects.
- Manage project status, priority and deadlines.
- Maintain team-member records.
- Provide project and team search.
- Establish relational database connections.
- Design a responsive Bootstrap interface.
- Validate user inputs and secure database operations.

1.5 Scope of the Project

Included in the system	Outside the current scope
User registration and login	Real-time chat and messaging
Personal project dashboard	File and document sharing
Project CRUD operations	Automated email notifications
Priority and status management	Third-party calendar synchronisation
Due-date validation	Automatic reminders for deadlines
Adding and removing team members	Role-based administrator panel
Session-based access	Mobile application

2. Modules with details

2.1 System Architecture

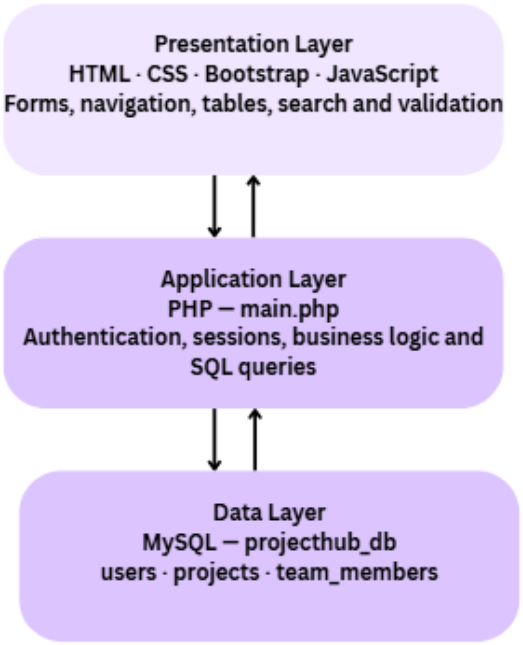


Figure 1: Three-layer logical architecture of ProjectHub

2.2 Technologies used

Software	Purpose
Windows or another compatible operating system	Development environment
XAMPP	Local Apache and MySQL/MariaDB environment
Apache	Web server
PHP	Server-side scripting and request processing
MySQL	Relational database management
phpMyAdmin	Database creation and administration
HTML5	Structure of web pages
CSS3	Styling and responsive layouts
Bootstrap 5.3.3	Responsive interface components
JavaScript	Client-side search and form validation
Visual Studio Code or equivalent	Source-code editing
Modern web browser	Application execution and testing

2.3 Functional Requirements

1. User registration and login.
2. Maintain authenticated user sessions.
3. Display project statistics and recent projects.
4. Add, view, edit and delete projects.

5. Manage project priority, status and deadlines.
6. Add and manage project team members.
7. Display a searchable team directory.
8. Validate forms.
9. Provide secure logout functionality.

2.4 MODULES

User Registration and Authentication Module

Purpose: To manage user identity and access.

Functions:

- User registration and login.
- Input validation.
- Secure password storage.
- Session management.
- Access control and logout.

Dashboard Module

Purpose: To provide an overview of user projects.

Features:

- Personalised greeting.
- Recent project details.
- Priority, status and deadline display.
- Quick project creation.
- Navigation to project management.

Project Management Module

Purpose: To display team members and their project associations.

Functions:

- Display member names and emails.
- Show associated project names.
- Identify project owners.
- Provide a centralised team directory.

Team Member Management Module

Purpose: To link team members with projects.

Functions:

- Select a project.
- Enter member details.
- Save member records.
- Associate members with projects.

Team Directory Module

Purpose: To display team members and their project associations.

Information Displayed:

- Member name and email.
- Associated project name.
- Project creator's name.

Due-Date Validation Module

Purpose: To ensure valid project deadlines.

Validation Steps:

- Check the date format.
- Reject past dates.
- Convert dates to MySQL format before saving.

Session Management Module

Purpose: To maintain user authentication across pages.

Working:

- Start or resume the PHP session.
- Store user details after login.
- Restrict unauthenticated access.
- Redirect users to login when required.
- Clear the session during logout.

3. Database design

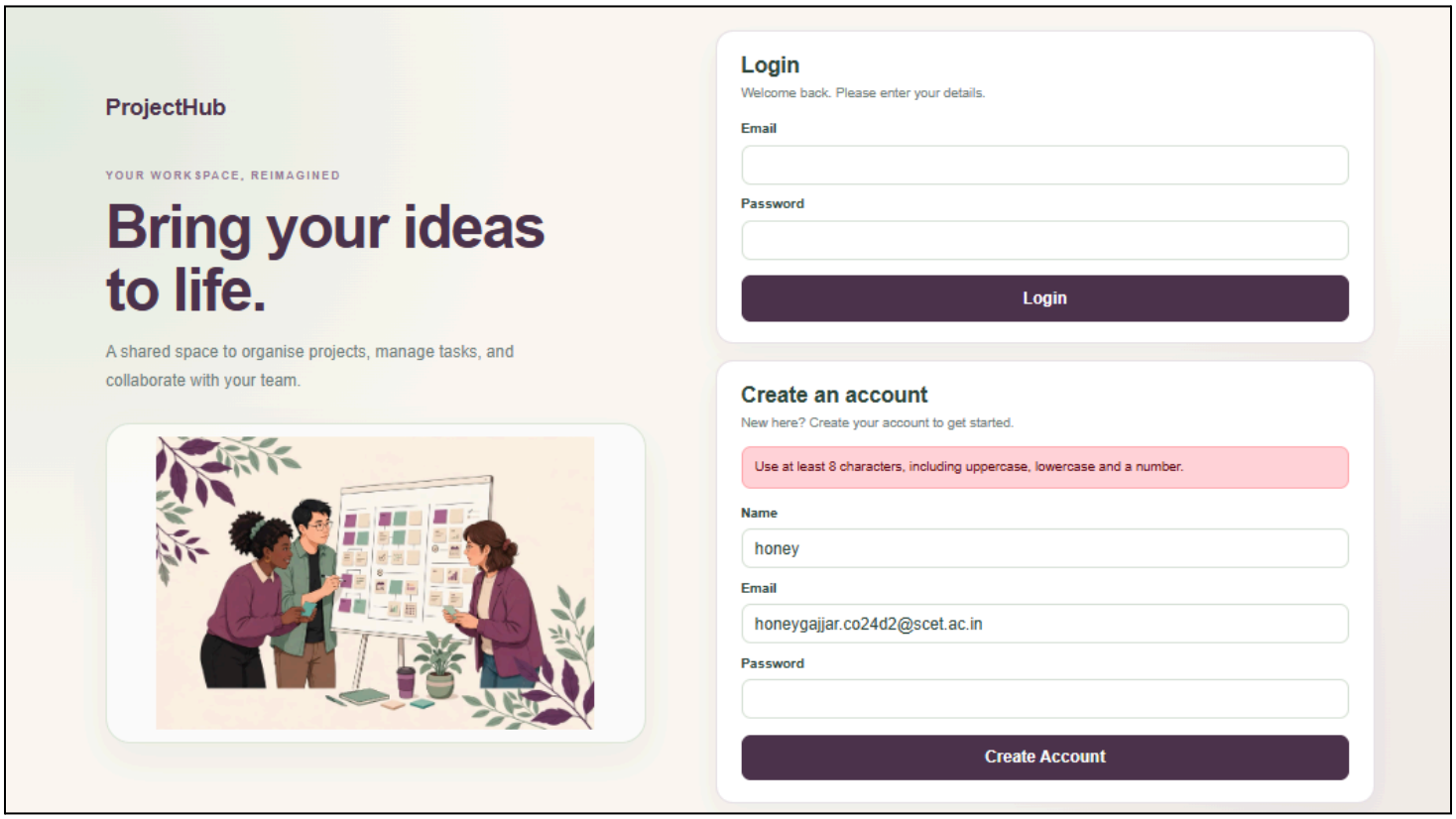
The project uses the MySQL database projecthub_db with three related tables:

- users: id (primary key), name, email (unique), password, created_at.
- projects: id (primary key), user_id (foreign key), project_name, description, priority, due_date, status, created_at.
- team_members: id (primary key), project_id (foreign key), name, email.

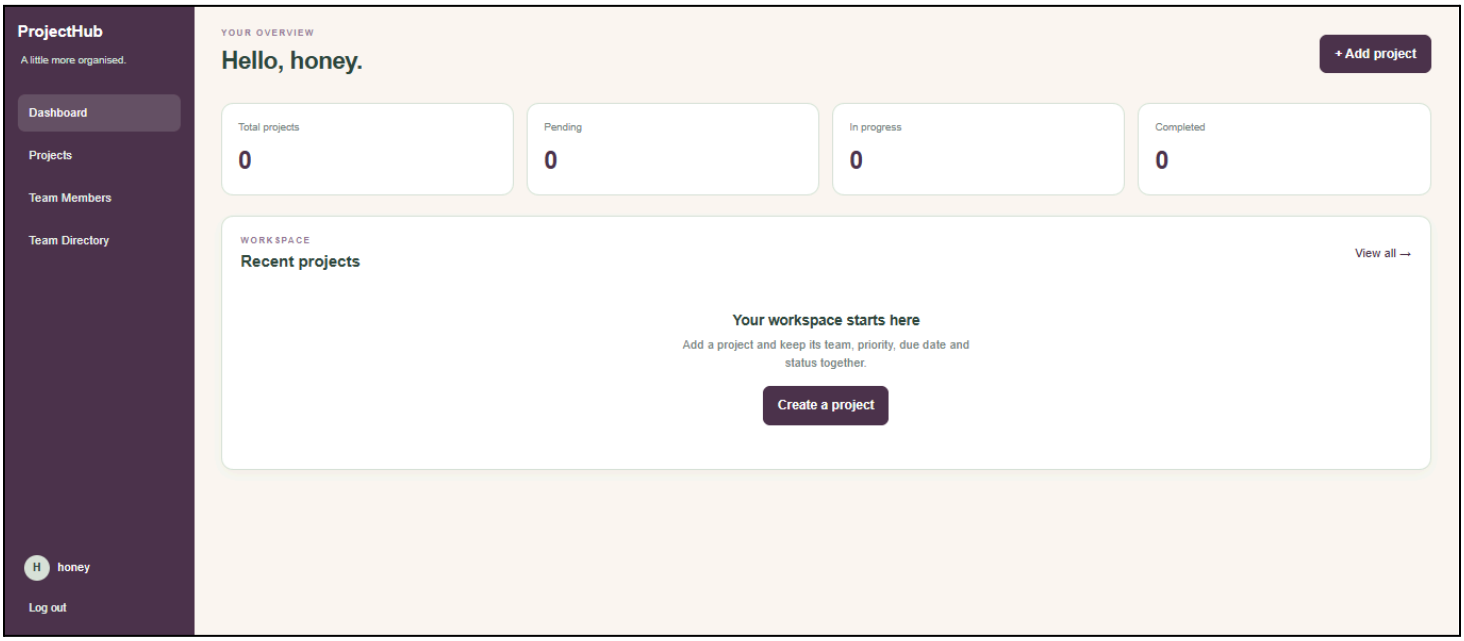
Relationships:

1. One user can own multiple projects (users 1:N projects).
2. One project can contain multiple team members (projects 1:N team_members).
3. Foreign keys maintain referential integrity; ON DELETE CASCADE removes dependent records when their parent record is deleted.

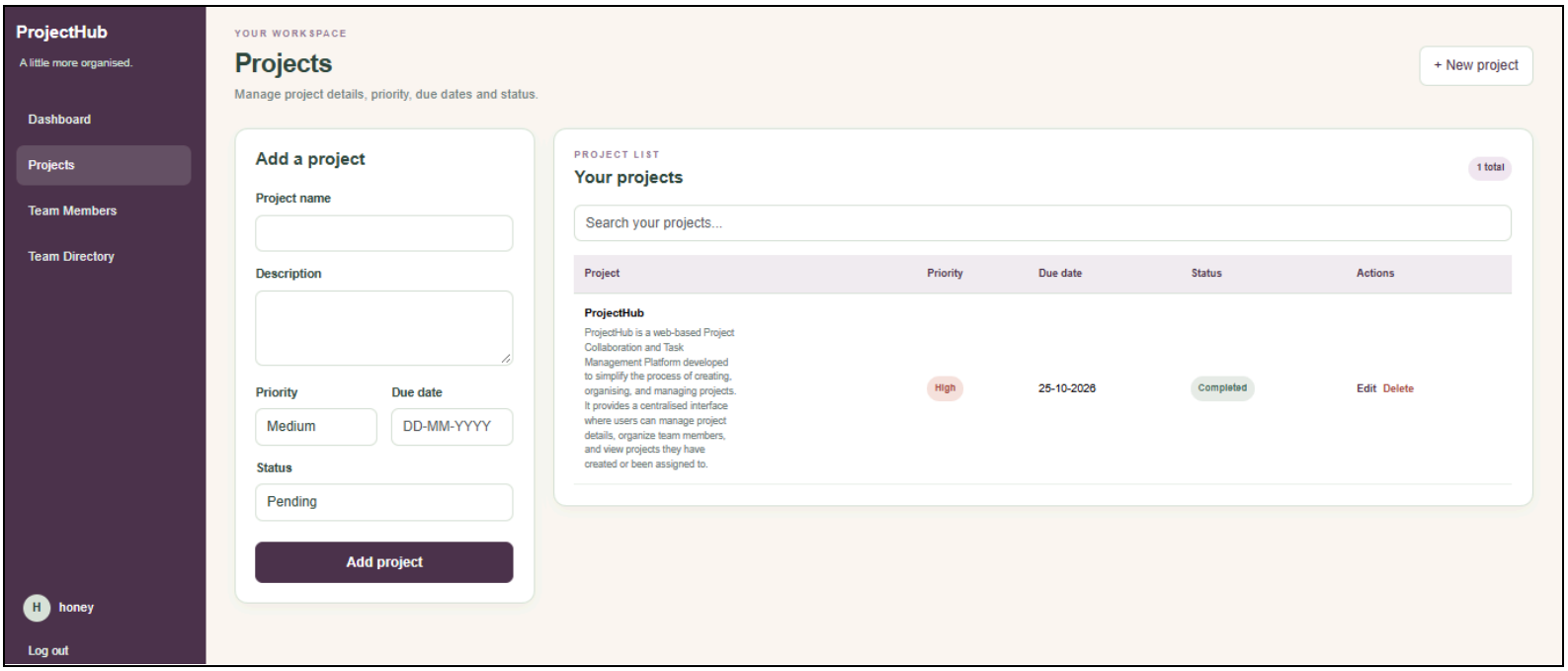
4. Screenshots



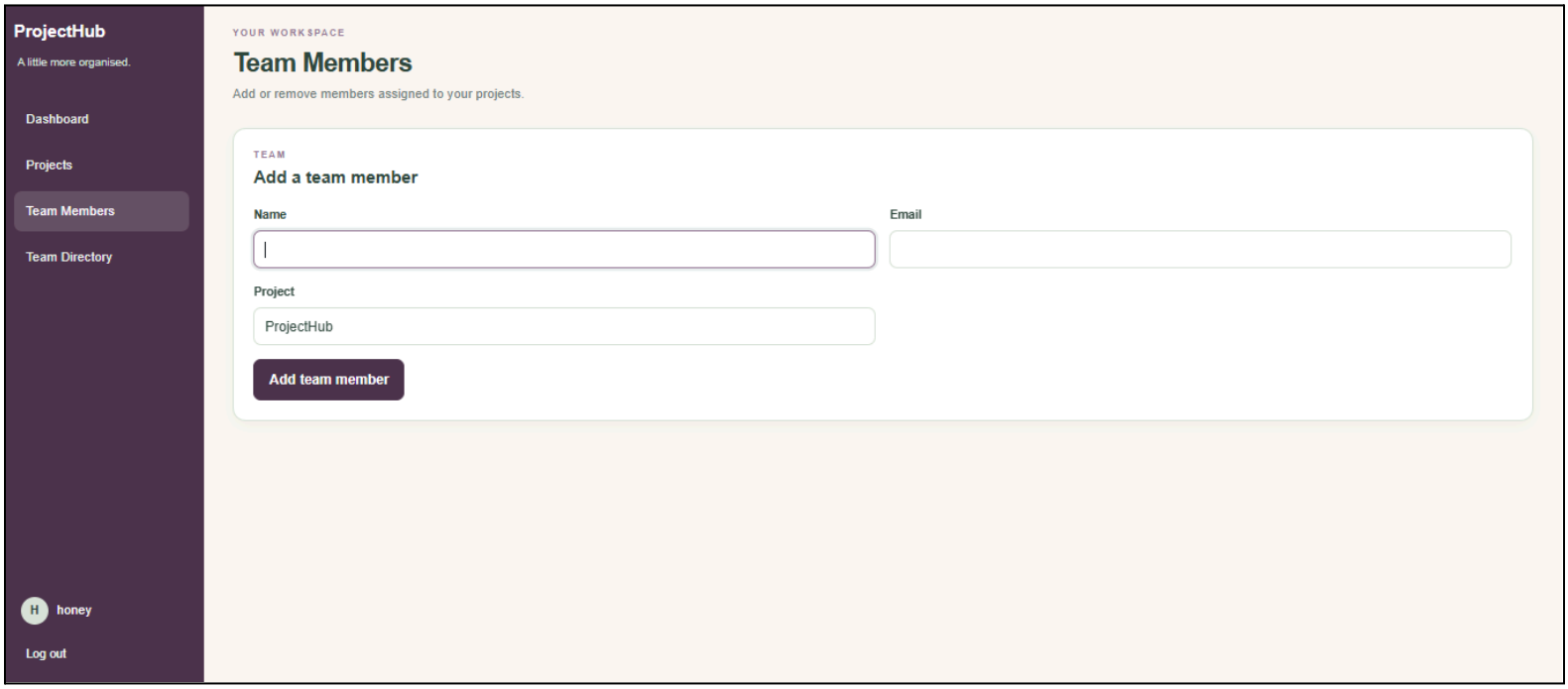
Screenshot 1: Login page



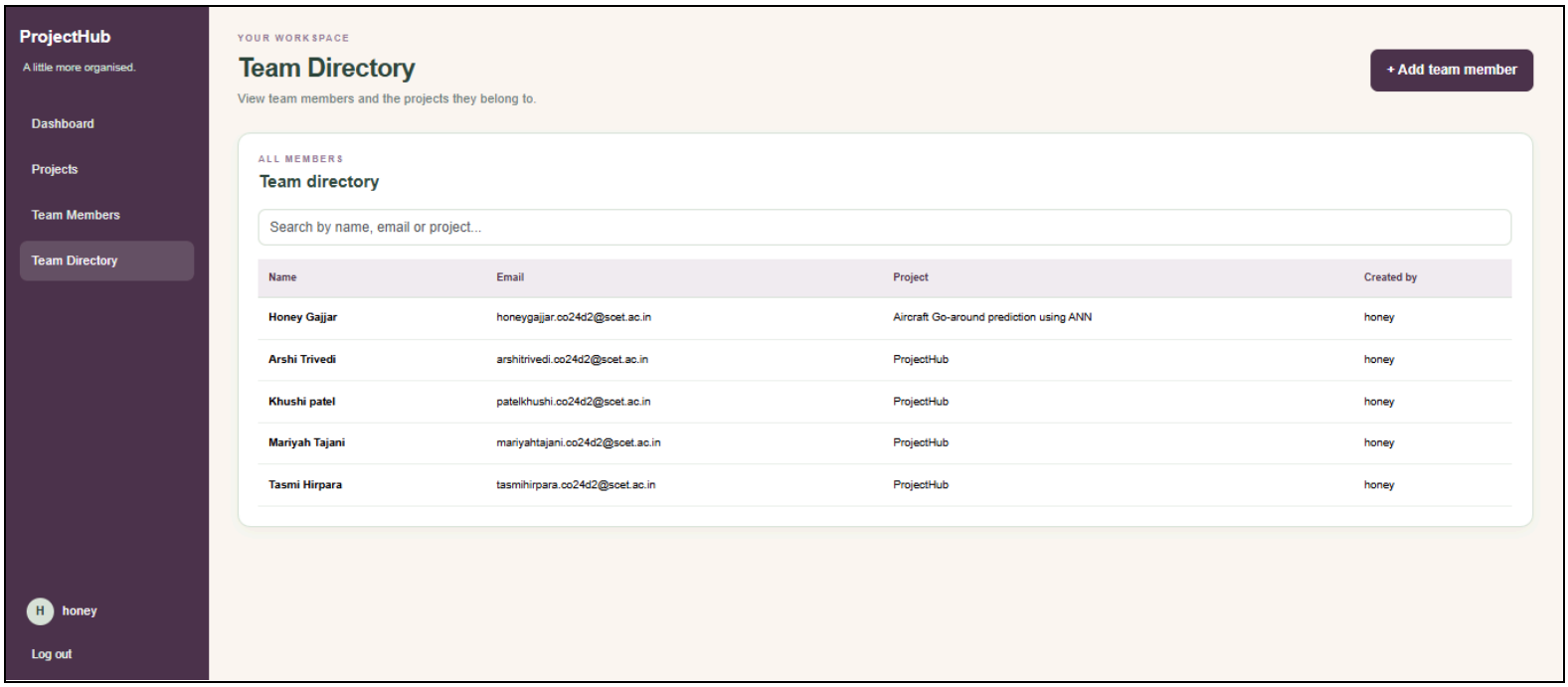
Screenshot 2: Initial dashboard while creating account



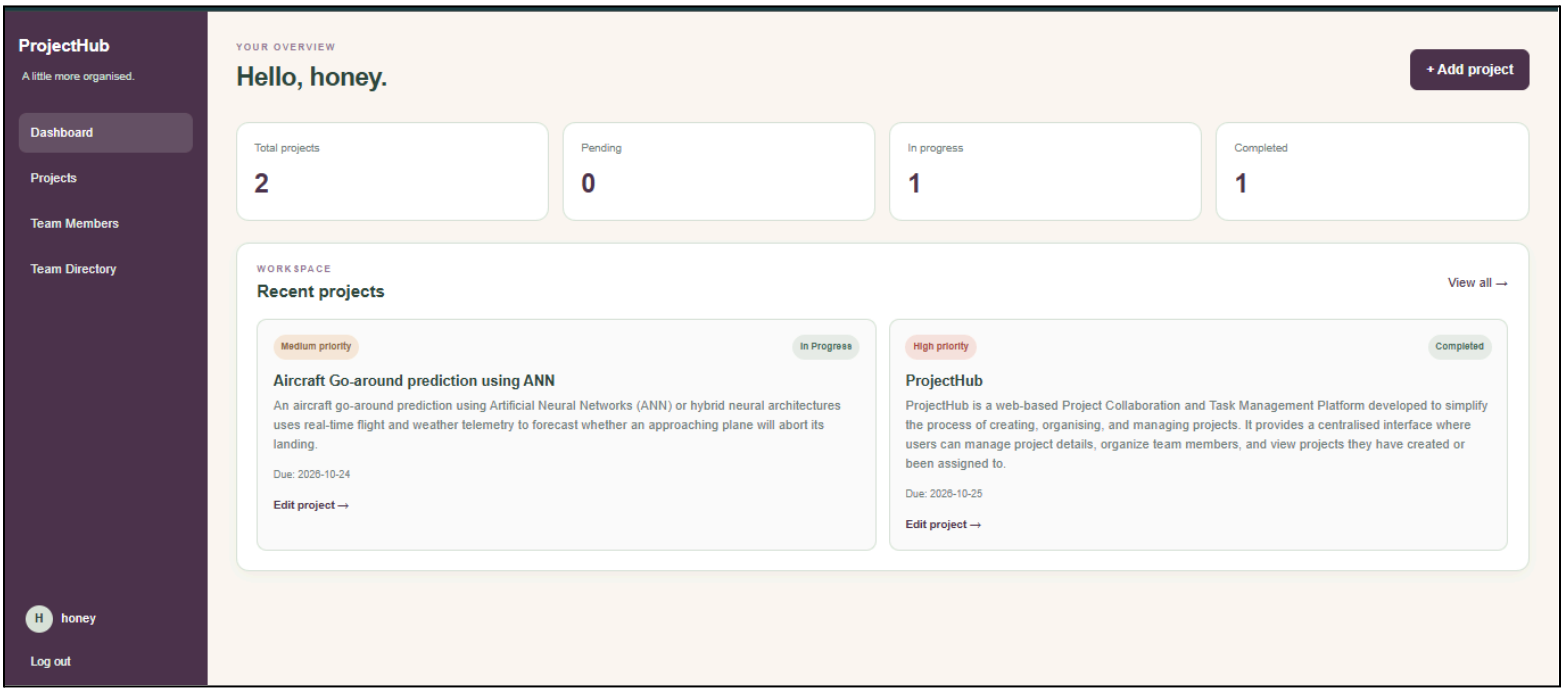
Screenshot 3: Project module page



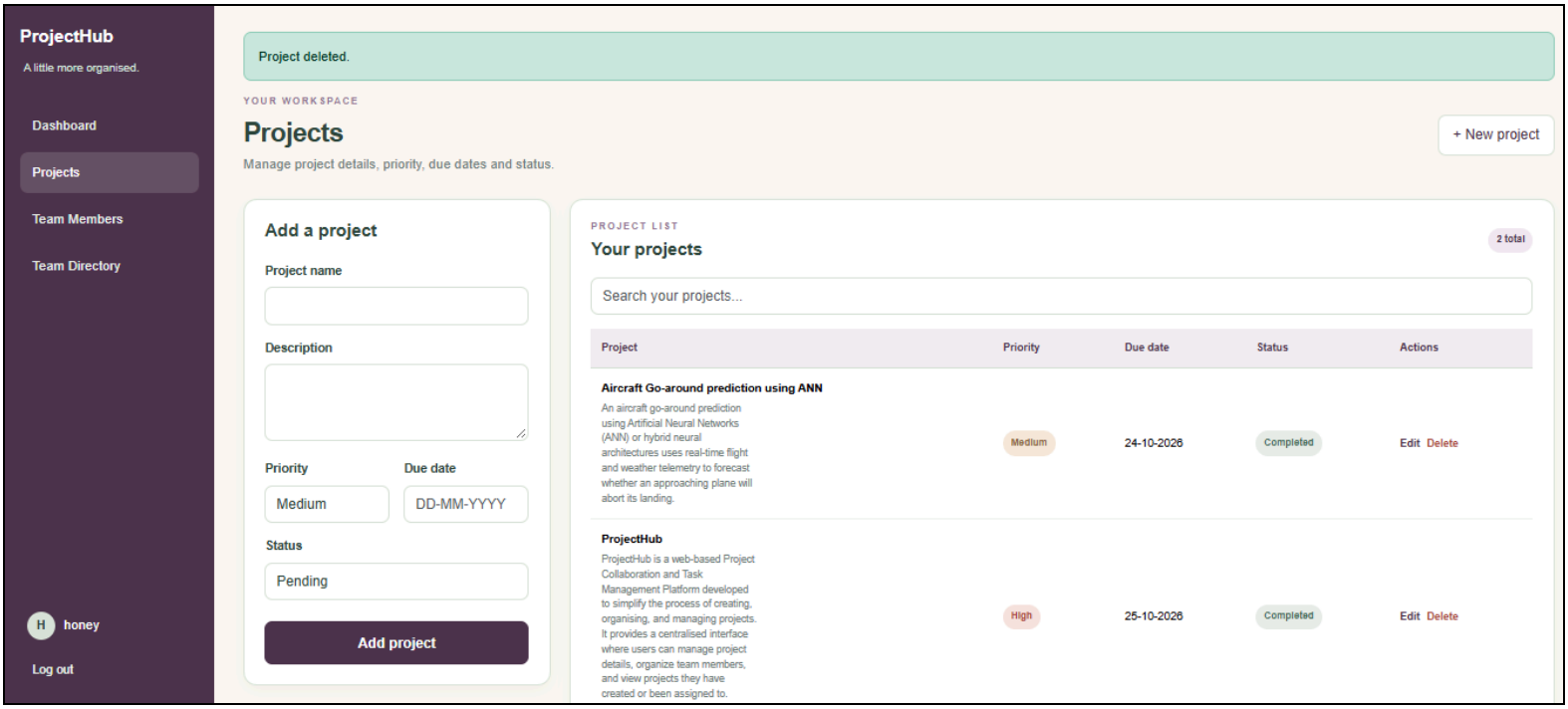
Screenshot 4: Team members page



Screenshot 5: Team directory



Screenshot 6: Dashboard while login



Screenshot 7: project editing (delete)

5. Conclusion

ProjectHub demonstrates a basic full-stack project-management application using PHP, MySQL, HTML, CSS, Bootstrap, and JavaScript. It centralizes project and team-member data and supports core record-management operations. Future improvements may include editable calendar scheduling, role-based permissions, notifications, and report export.

6. References

PHP Documentation — https://www.php.net/docs.php

MySQL Reference Manual — https://dev.mysql.com/doc/

Bootstrap Documentation — https://getbootstrap.com/docs/

MDN Web Docs — https://developer.mozilla.org/

Apache Friends (XAMPP) — https://www.apachefriends.org/